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Phylogenetic Relationships of Aracanin Genera of Boxfishes (Ostraciidae: Tetraodontiformes)

RICHARD WINTERBOTTOM AND JAMES C. TYLER

A cladistic analysis of the external characters, osteology and myology of the seven genera currently assigned to the ostraciid (trunkfishes and boxfishes) subfamily Aracaninae (boxfishes) results in our proposal of its division into two tribes, the Aracanini (*Aracana, Strophiurichthys, Anoplocapros, Caprichthys* and *Capropygia*) and the Kentrocaprini (*Kentrocapros* and *Polyplacapros*). In the case of the Aracanini, the phylogeny can be derived from the classification either from sequencing or from the numerical suffix method. Evidence is presented which corroborates the hypotheses that: a) the Aracaninae is monophyletic; b) the Ostraciinae is monophyletic; c) the Aracaninae and Ostraciinae are sister groups (the Ostraciidae); d) the Balistidae is monophyletic; and e) the Balistidae and Ostraciidae are sister groups.

OUR interest in the generic interrelationships of aracanin ostraciids was renewed by the recent description of a new genus and species of that subfamily, *Polyplacapros tyleri* Fujii and Uyeno. The new taxon is unique in having the caudal peduncle nearly completely covered with discrete bony plates (= modified scales) and in its almost fusiform body shape (Fig. 1), characters considered by Fujii and Uyeno (1979) to be primitive for ostraciids generally. This could, from a phylogenetic viewpoint, imply: a) that *Polyplacapros* may be the sister group of all other ostraciids; b) that it could be the sister group of all other aracanins; or c) that it could be the sister group of all other ostraciins. Fujii and Uyeno (1979) point out that the enlarged supraneural and expanded plate-like postcleithrum (Fig. 2) of *Polyplacapros* indicate that it is an aracanin (since these are specializations) and, if their analysis of character polarity of the peduncle scales and body shape is correct, then *Polyplacapros* is the sister group of all other aracanins. However, they also state that, among aracanin genera, ". . . *Polyplacapros*, . . . , appears to be closest to the genus *Kentrocapros* . . ." (Fujii and Uyeno, 1979:9). In terms of phylogenetic reasoning, there is thus a contradiction—*Polyplacapros* cannot be both the sister group of all other aracanins and also the sister group of *Kentrocapros* alone. This study was undertaken to resolve that contradiction, as well as to elucidate the phylogeny of aracanin genera (as they are presently understood).

Historical surveys of the literature pertinent to this topic may be found in Winterbottom (1974a, myology only) and Tyler (1980, osteology and external features). Winterbottom dissected representatives of only two of the seven currently recognized genera and made no comment on aracanin interrelationships, but Tyler (1980: Fig. 146) provided a diagram of his interpretation of the phylogenetic relationships of six of the seven aracanin genera (i.e., exclusive of *Polyplacapros,* the description of which was not yet available for discussion). As stated in his monograph, Tyler's preference for methods of phylogenetic reconstruction at that time was gradistic, i.e., ". . . to weigh and balance the value of both the generalized and specialized features . . ." (Tyler, 1980:28). The present study is based on the belief that phylogenetic relationships can be revealed most logically only by the presence of shared specialized characters (synapomorphies), thus following the principles outlined by Hennig (1966).

For the purposes of this study, we follow Tyler (1980:195, footnote) and Matsuura and Yamakawa (in press) in regarding *Aracanostracion* Smith 1949 as a synonym of *Kentrocapros.* Winterbottom's (1974a) classification will be followed; it places the Aracaninae and Ostraciinae in the Ostraciidae and includes balistins and monacanthins in the Balistidae, these two families being united in the Balistoidea. Tyler (1980) recognized each of these taxa at the next higher level and does not abandon that view here.

For any study purporting to use phylogenetic methodology, comparisons can best be made under the following conditions: a) the taxon being analysed is monophyletic; b) the monophyletic sister group of the taxon being analysed has been postulated; and c) the monophy-

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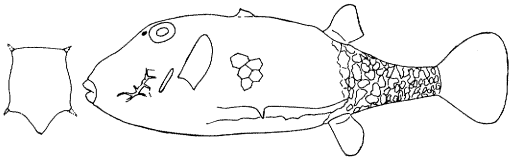


Fig. 1. *Polyplacapros tyleri*, left lateral view (right) and cross-section of body (left).

letic sister group of these two groups has been postulated. Thus, analysis is performed using the outgroup comparison method.

In Winterbottom's (1974a:93) analysis, myological evidence is presented which corroborates the hypotheses that: a) the aracanins and ostraciins are sister groups, forming the Ostraciidae (with 36 autapomorphies); b) the sister group of the ostraciids, the Balistidae, is monophyletic (with 16 autapomorphies); and c) the balistids plus ostraciids, the Balistoidea, is monophyletic (with 11 autapomorphies). However, he presented no autapomorphies to define either the Aracaninae or the Ostraciinae as monophyletic. Winterbottom's phylogenetic tree (1974a: Fig. 47) for this section of the extant tetraodontiforms is identical to Tyler's (1980: Figs. 1, 2) groupings of these taxa.

Tyler's (1980) osteological diagnoses of the taxa involved (Balistoidea, Balistidae, Ostraciidae, Aracaninae and Ostraciinae, in the terminology used here) are a mixture of primitive and derived character states. Therefore, we have extracted from his diagnoses those character states that we believe to be derived for each of the above levels of universality and we have added derived myological characters diagnosing both the aracanins and the ostraciins from the data presented by Winterbottom (1974a). We also note that the characters given by Tyler (1980) for his Balistoidei (a group which, in addition to the Balistoidea of this paper, includes the triacanthoid tetraodontiforms) are all primitive.

We here regard the Eocene *Eoplectus* as the most generalized tetraodontoid so far discovered. It shares with all tetraodontoids the synapomorphy of a crushing "beak" with the teeth of the jaws intimately incorporated into the dentaries and premaxillaries. Plesiomorphous characters retained by *Eoplectus* include a spiny dorsal fin and an I 4 pelvic fin. For the Balistoidea, we use as outgroups the two (different) sister groups which have been proposed for these fishes, the triacanthodids plus triacanthids (Tyler, 1968, 1980), and the Tetraodontoidea (Winterbottom, 1974a). It is not our purpose here to evaluate these two hypotheses, nor to further explore Winterbottom's (1974a:95–96) innuendo that the Acanthuridae may be more intimately associated with balistoids and tetraodontoids than with either the triacanthodids or triacanthids. However, we do comment on character states in acanthurids where we feel it is appropriate, although the possible outgroup role of acanthurids has not been documented.

Materials and Methods

Methods are as outlined in Winterbottom (1974b) for myology and in Tyler (1980) for osteology. Material examined is as listed in those two publications, plus the following additional specimens examined for this work (ANSP = Academy of Natural Sciences of Philadelphia; AMS = Australian Museum, Sydney):

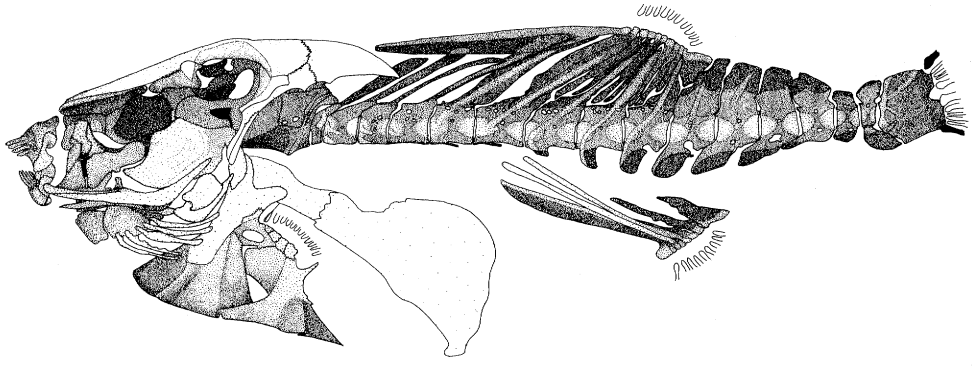


Fig. 2. *Polyplacapros tyleri*, left lateral view of entire skeleton.

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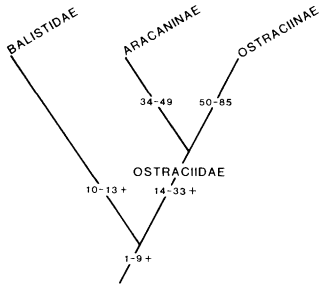


Fig. 3. Phylogenetic relationships of the Balistoidea. Numbers refer to the autapomorphies as listed in the text. A "plus" sign after the numbers indicates that additional autapomorphies, not listed here, in the myology of the taxa can be found in Winterbottom (1974a:93).

Anoplocapros amygdaloides, 103.7 mm SL, Western Australia, ANSP 114116.
Aracana aurita, 84.5 mm SL, Western Australia, ANSP 112697.
Caprichthys gymnura, 66.1 mm SL, Western Australia, ANSP 114117.
Polyplacapros tyleri, 256 mm SL, off New South Wales, AMS I. 21447-001.
Strophiurichthys robustus, 155.6 mm SL, New South Wales, ANSP 109567.

Detailed descriptions of the myology and osteology are not repeated here, since they are available in Winterbottom (1974a) and in Tyler (1980), respectively. Most comparisons can be made using the figures presented in those two papers and those herein.

To prevent possible confusion, it should be borne in mind that we follow the definitions of Hennig (1966) in that an autapomorphy is a specialized character shared by the members of a given taxon, a synapomorphy is a specialized character shared by two or more taxa. If the taxa in the latter case are monophyletic, and one uses their all-encompassing name, the "synapomorphy" becomes an "autapomorphy" at that higher level of application (e.g., a bony cuirass of modified scales is an autapomorphy for the Ostraciidae, but a synapomorphy for the Aracaninae and Ostraciinae). We give this detailed explanation because we have frequently seen and heard these words used in ways other than those proposed by Hennig (1966:90), a practice which, we feel, merely creates confusion.

Diagnostic Characters of the Higher Taxa

Osteological autapomorphies for the Balistoidea (with the plesiomorphous states in parentheses) are: 1) loss of the ascending process or pedicel of the premaxilla (vs well developed as in triacanthoids or reduced but still present as in tetraodontoids); 2) interdigitation of the maxilla with the premaxilla (vs maxilla moveably articulated with premaxilla in triacanthoids; however, this may be an autapomorphy for balistoids plus tetraodontoids and possibly also plus acanthurids, i.e., it might define a more inclusive group—until this possibility is investigated further, we use it as above); 3) anterolateral margin of ethmoid expanded dorsolaterally (vs curving ventrolaterally from the midline in triacanthoids, with a lesser degree of such dorsolateral expansion evident in generalized tetraodontoids and acanthurids); 4) a well developed anteroventral shelf of bone from the prootic over the ventrolateral floor of the orbit (vs no well developed prootic shelf); 5) parasphenoid a broad, flat, vertical plate of bone anterior to the orbit (vs no plate, although the preorbital section of the parasphenoid is deeper in generalized tetraodontoids and at least some acanthurids than in triacanthoids); 6) hyomandibular articulating only with the prootic and pterotic, to the exclusion of the sphenotic (vs all three of these bones contributing to the articulating fossa of the hyomandibular); 7) interoperculum a short, rod-like bone, not extending posteriorly beyond the junction of the epi- and inter-hyal (vs extending posteriorly well beyond this point, except in most tetraodontoid families where it is similar to, or a little longer than, the balistoid condition); 8) well developed teeth, where present, on pharyngobranchials (=infrapharyngobranchials) 2 or 3 only (vs teeth present on pharyngobranchials 2 through 4 in at least the generalized members of the outgroups); 9) pelvic fin reduced to minute "fin-ray elements" (Tyler, 1962, 1980) or absent altogether (vs at least well developed pelvic spines usually followed by 2 to 5 pelvic-fin rays in generalized triacanthoids and in *Eoplectus*, as well as in acanthurids).

To these nine balistoid autapomorphies may be added the 11 myological ones listed by Winterbottom (1974a:93), which need not be repeated here.

Osteological autapomorphies for the Balistidae are: 10) palatine either a "T" shaped bone

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with the "foot" of the shaft articulating with the ectopterygoid, or represented by the head of the "T" with a cord-like ligament to the ectopterygoid representing the shaft (vs an oblong plate of bone intimately connected to the ecto- and meso-pterygoids); 11) first pterygiophore of spiny dorsal fin attached to the chondrocranium (vs positioned posteriorly and not in contact with skull in ostraciids, with both conditions apparently apomorphic and derived from a condition where the spinous dorsal fin lies close behind but not in contact with the skull, as is found in *Eoplectus*, triacanthoids and acanthurids); 12) first dorsal spine capable of being locked in position by the "jamming" action of the second dorsal-fin spine (vs spiny dorsal fin autapomorphically absent in ostraciids, and no such locking mechanism in the spiny dorsal fin of the further-removed outgroups); 13) pelvis an elongate rod of bone with, primitively, a dorsal flange on its posterodorsal surface (vs autapomorphically absent in ostraciids, and of more usual perciform shape in the more generalized members of the other outgroups).

To these four osteological autapomorphies of balistids, we add the 16 myological ones listed by Winterbottom (1974a:93).

We realize that the balistid pelvis is of generally similar appearance to that of the tetraodontoid *Triodon* (the only representative of the most generalized extant family of tetraodontoids and the sister group of the tetraodontids, diodontids and molids) but conclude that this is due to convergence because of: a) the autapomorphies listed above for balistoids; b) the 15 myological characters of tetraodontoids listed by Winterbottom (1974a:93) and c) the presence of a relatively primitive pelvis and a pelvic fin (one spine and four rays) in the sister group of the extant tetraodontoids, the Eocene *Eoplectus*.

Osteological autapomorphies for the Ostraciidae (with the plesiomorphous condition found in balistids and, where applicable, the more generalized outgroups, in parentheses) are: 14) a prominent ventral or ventrolateral flange on the lateral margin of the prootic shelf (vs no flange); 15 myodome small or absent (vs well developed); 16) parasphenoid with a dorsal flange in the orbit sutured to anteroventral extensions of the pterosphenoids (vs no contact between these two bones); 17) hyomandibular a broad, flat plate of bone (vs a more or less flattened shaft); 18) distal end of posterior branchiostegal ray articulated with medial face of suboperculum (vs lying free in membrane); 19) urohyal very small, with a vestigial or no ventral flange (vs well developed with a prominent ventral flange); 20) an anteroventromedial flange on the cleithrum (vs no flange); 21) Baudelot's ligament ossified (vs unossified); 22) coracoid expanded ventrally, as wide as or wider there than dorsally, with prominent, broad lateral flanges (vs coracoid narrower ventrally than dorsally and without lateral flanges); 23) actinosts (=pectoral radials) immovably sutured together and to the scapula or coracoid (vs not sutured); 24) loss of the pelvis and pelvic fins (vs pelvis present and a vestigial pelvic fin present in generalized and most other balistids); 25) at least the anterior two vertebrae reduced in

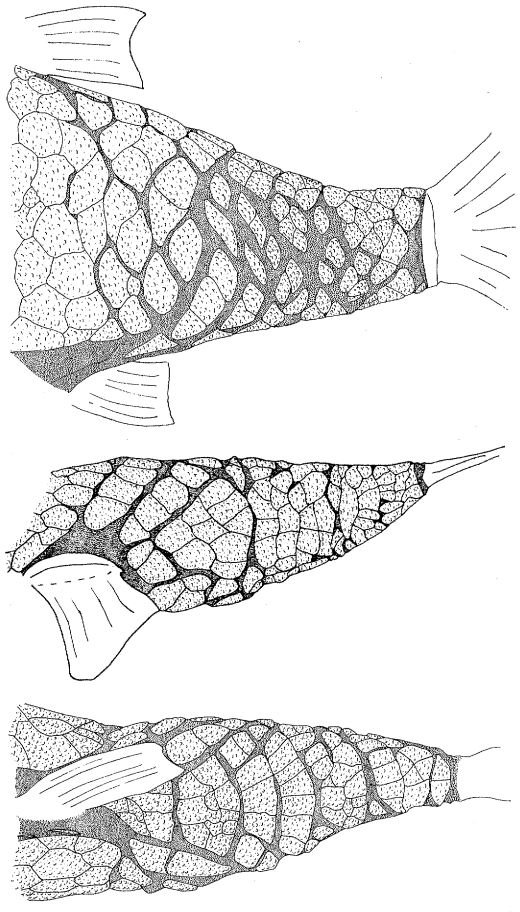


Fig. 4. *Polyplacapros tyleri*, caudal peduncle: (top), left lateral view; (middle), dorsal view; (bottom), ventral view.

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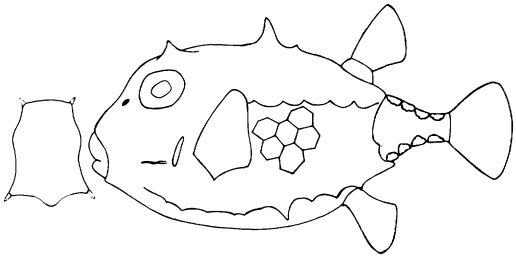


Fig. 5. *Kentrocapros aculeatus*, left lateral view (right) and cross-section of body (left).

size and partially fused to each other and to the rear of the skull (vs these vertebrae of normal size and not fused); 26) vertebrae following the fused anterior two vertebrae with anteroposteriorly compressed centra (vs not compressed, while the compressed vertebrae in ostraciins may be fused to the anterior two vertebrae); 27) epipleural ribs absent (vs present); 28) all elements of the caudal-fin supporting structures that are derived from or associated with pu I (ultimate centrum) and the compound ural centrum either fused into a single plate or absent (vs composed of four separate elements); 29) 10–11 caudal-fin rays (vs 12); 30) spiny dorsal fin absent (vs anteriorly placed reduced spiny dorsal fin—see autapomorph 11); 31) dorsal and anal fins without distal pterygiophores (vs at least the anterior region of these fins with distal pterygiophores); 32) anterior four to six anal pterygiophores with their distal tips in the midline but their proximal ends diverging variously to the left or right of the midline (vs all pterygiophores with their proximal ends in the midline); 33) head and most of the body encased in cuirass of hexagonal scale plates whose edges interdigitate (vs discrete, albeit modified, non-interdigitating scales).

To these 20 derived osteological characters of ostraciids may be added the 36 myological ones for this taxon listed by Winterbottom (1974a:93).

Osteological autapomorphies for the Aracaninae (with the plesiomorphous condition in the ostraciin and balistid outgroups in parentheses) are: 34) epiotic (=epioccipital) with a strong posterolateral process (vs no process); 35) a long, vertical crest extending posteriorly from the supraoccipital (vs no crest as in ostraciins or a small vertical crest as in balistids); 36) prootic shelf with a recurved wing of bone from its anterolateral edge which passes posteriorly to contact the anterior edge of the laterally expanded posterior region of that bone (vs no lateral strut); 37) a broad suture in the interorbital septum between the parasphenoid and pterosphenoids (vs a narrow suture as in ostraciins, or no contact at all as in balistids); 38) frontals extending anteriorly over the ethmoids to or almost to the anterior tip of the skull (vs ending anteriorly in the vicinity of the prefrontals); 39) ventral postcleithrum tremendously expanded as a broad, thin plate of bone covering much of the lateral surface of the abdominal cavity [vs short and not expanded as in ostraciins, or strut-like as in balistids, when present as a separate element (A new genus and species of balistid, *Xenobalistes tumidipectoris*, recently described by K. Matsuura (1981) has a much expanded postcleithrum which contacts the coracoid anteriorly. It shares the four osteological autapomorphies of balistids listed above, while it has none of the 19 osteological autapomorphies of ostraciids mentioned above. Therefore, we conclude that the expanded condition of the ventral postcleithrum in this new genus of balistid and in aracanins is due to convergence.)]; 40) upper pectoral-fin ray of two equal-sized halves (vs a single piece representing the medial half or a larger medial half and a much smaller lateral nubbin of bone. We note here that the presence of subequal halves of the upper pectoral-fin ray is the usual condition for teleosts, but point out that our two outgroups, ostraciins and balistids, have the lateral half reduced or absent. We conclude that this must have been the condition in the common ancestor of aracanins and ostraciins, and that the redevelopment of equal-sized halves in the upper pectoral-fin ray of aracanins is autapomorphic); 41) well developed transverse processes on the anterior five abdominal vertebrae (vs poorly developed or absent).

Derived myological characters for the Aracaninae are: 42) dilatator operculi a strap-like, thin muscle in which the fibers are almost parallel (vs a more developed bipinnate muscle); 43) pharyngoclavicularis externus very well developed (vs almost half that size in balistids and ostraciins); 44) obliquus ventralis III poorly developed (vs about twice that size in balistids and ostraciins); 45) posterodorsal fibers of levator internus III visible beneath the pterotic in superficial lateral view (vs covered from lateral view by the pterotic); 46) levator internus III with two insertional tendons, one to pharyn-

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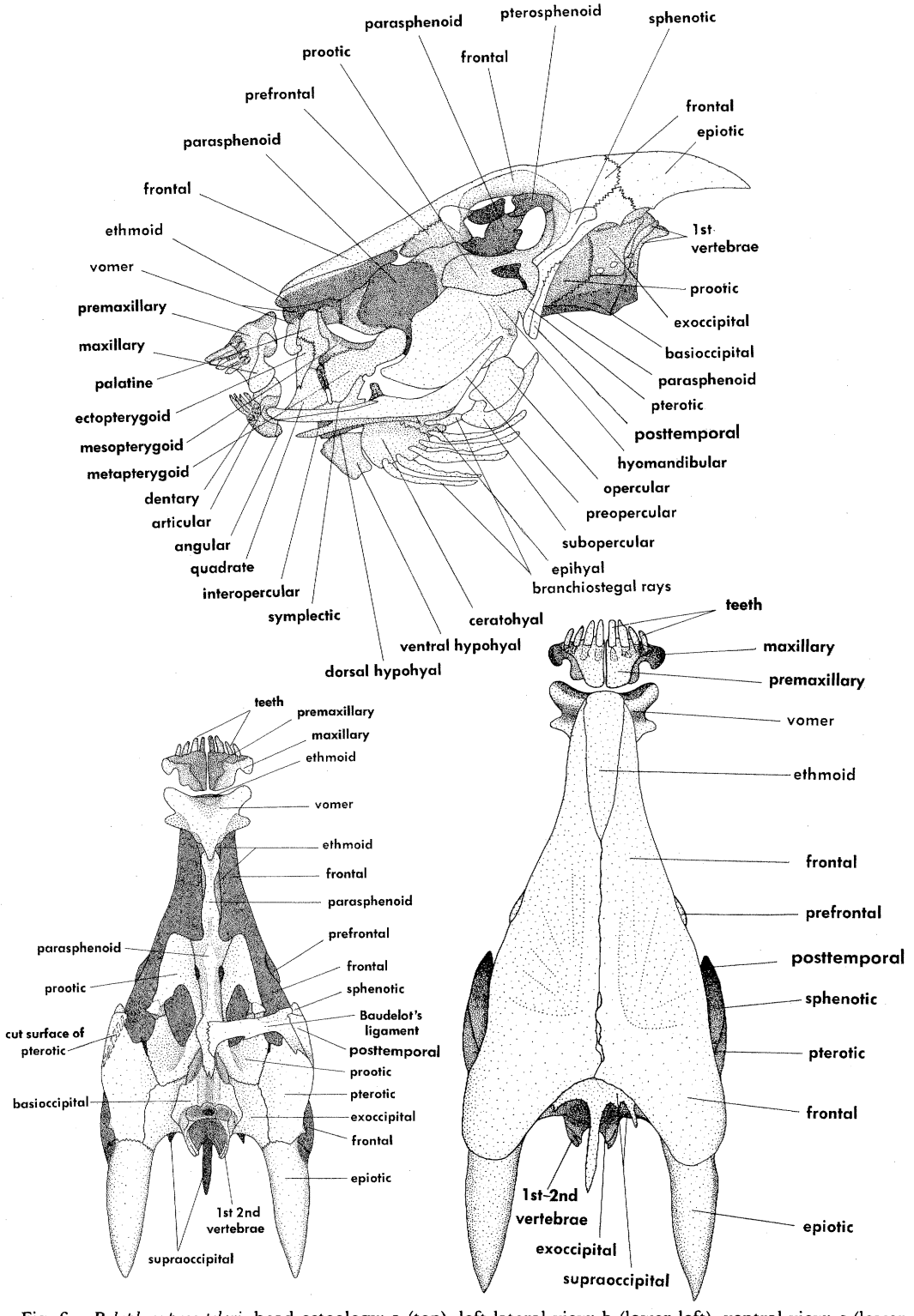


Fig. 6. *Polyplacapros tyleri*, head osteology: a (top), left lateral view; b (lower left), ventral view; c (lower right), dorsal view.

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gobranchial 3 and the other to epi- and pharyngobranchial 4 (vs a single tendon to pharyngobranchial 3 only); 47) loss of transversus dorsalis III (vs present); 48) a pair of ligaments from the ventral surface of the sphincter oesophagi to the medial faces of the cleithra (vs no ligaments); and 49) erector dorsalis lies lateral to adductor profundus (vs between the adductor profundus and adductor superficialis).

Osteological autapomorphies for the Ostraciinae are: 50) anterior region of ethmoid expanded laterally into a broad plate (vs ethmoid not so expanded anteriorly, of more or less continuous width); 51) ventral margin of parasphenoid expanded laterally anterior to its regions of suturing to the anteromedial edges of the prootic shelves, forming a "hard palate" over the roof of the oral cavity (vs not laterally expanded in this region, or, if so, not as extensively); 52) no crest at all from the posterodorsal margin of the supraoccipital (vs the supraoccipital crest small as in balistids or well developed as in aracanins); 53) myodome very small or absent (vs developed as usual for aracanins and balistids); 54) posterior process from posterodorsal end of coracoid absent (vs process present); 55) ventral postcleithrum small or absent (vs enormously expanded in aracanins and strutlike in balistids); 56) a total of four to five anterior vertebrae variously fused and sutured to each other and to the occipital region of the skull (vs two as in aracanins or no anterior vertebral fusion as in balistids); 57) a prominent ventral process from the compound centrum of the fused vertebrae (vs no process in this area); 58) vertebra following fused anterior vertebrae with a short neural spine (vs neural spine larger); 59) no transverse process on the vertebra following the fused vertebrae (vs a transverse process present); 60) at least some vertebrae (in addition to the anterior four or five) sutured to each other, usually between the seventh abdominal and the first caudal vertebrae (vs no additional vertebral fusions); 61) hemal spines of anterior caudal vertebrae poorly developed or absent (vs hemal spine larger); 62) hemal canal diverted to the left or right under the centra of the more posterior abdominal vertebrae (vs hemal canal lying in the midline); 63) centra of at least two of the vertebrae preceding the caudal plate much compressed anteroposteriorly (vs not so compressed); 64) hemal spine of second pleural centrum poorly developed (vs hemal spine larger); 65) basal pterygiophore which formerly supported the spiny dorsal fin short (vs very long in aracanins, where it is possibly also autapomorphic, or bearing spines and moderately long, as in balistids); 66) distal end of first basal pterygiophore of anal fin laterally expanded (vs not laterally expanded); 67) 10 caudal-fin rays (vs 11 in aracanins and 12 in balistids); 68) carapace closed behind anal fin (vs carapace open behind anal fin); and 69) caudal peduncle without isolated scale plates (vs isolated scale plates present).

Myological autapomorphies for the Ostraciinae are: 70) dorsolateral portion of A2α lies lateral to ventrolateral fibers of A2β (vs lying medial to A2α); 71) retractor arcus palatini inserts via consolidated tendon (vs a musculous insertion); 72) sternobranchialis consists of two muscles (vs a single muscle); 73) origin of anterior fibers of sternobranchialis extends posteroventrally to the fascia overlying the pectoral fin abductors and the anteroventrolateral face of the cleithrum below this (vs no extension of sites of origin for the sternobranchialis); 74) obliquus dorsalis III absent (vs present); 75) transversus dorsalis I present (vs absent); 76) retractor dorsalis originates from the lateral faces of the first few vertebrae (vs from lateral surface of basioccipital); 77) loss of adductor IV (vs present); 78) loss of coracoradialis (vs present); 79) loss of supracarinalis (vs present); 80) loss of flexor dorsalis (vs present); 81) loss of flexor ventralis (vs present); 82) epaxialis attached to vertebral column only posterior to dorsal fin (vs attached medially to vertebral column throughout its length); 83) epaxialis and obliquus inferioris unite posteriorly to form a cruciform tendon (vs the two muscles remain separate); 84) loss of the anterodorsal portion of the obliquus inferioris which connects the epiotic (in aracanins) or midlateral septum (balistids) to the dorsal margins of the postcleithra (vs present); and 85) development of a complex bundle of muscle fibers on the anterodorsal margin of the swimbladder (vs swimbladder muscles absent).

In summary, 20 autapomorphies define the Balistoidea (nine listed here plus 11 from Winterbottom, 1974a); 20 the Balistidae (four listed here plus 16 from Winterbottom, 1974a); 56 the Ostraciidae (20 listed here plus 36 from Winterbottom, 1974a); 16 the Aracaninae (all listed here) and 36 the Ostraciinae (all listed here) (Fig. 3). We believe that this degree of corroboration justifies our selection of the os-

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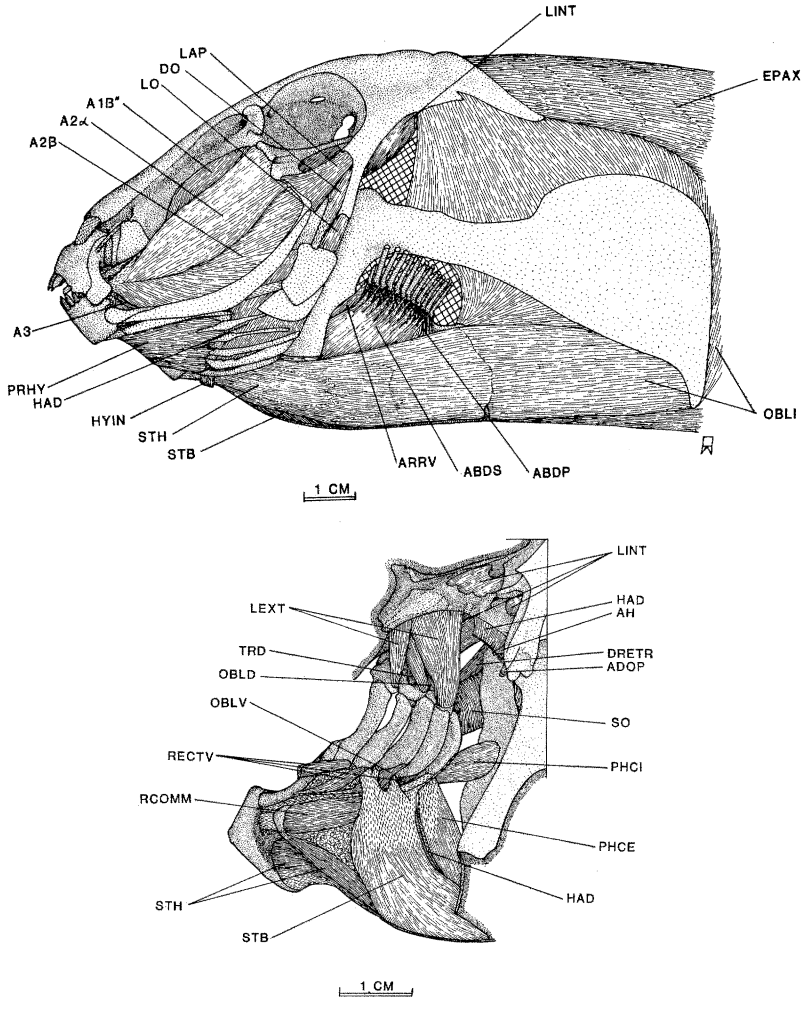


Fig. 7. *Polyplacapros tyleri*: (top), left lateral view of superficial anterior musculature; (bottom), left lateral view of superficial branchial musculature. Figure abbreviations: A1. = Adductor mandibulae, A1β''. A2. = Adductor mandibulae, A2α and A2β. A3. = Adductor mandibulae, A3. ABDP. = Abductor profundus. ABDS. = Abductor superficialis. ADOP. = Adductor operculi. AH. = Adductor hyomandibulae. ARRV. = Arrector ventralis. DO. = Dilatator operculi. DRETR. = Retractor dorsalis. EPAX. = Epaxialis. HAB. = Hyohyoidei abductores. HAD. = Hyohyoidei adductores. HYIN. = Hyohyoideus inferioris. LAP. = Levator arcus palatini. LEXT. = Levatores externi. LINT. = Levatores interni. LO. = Levator operculi. OBLD. = Obliquus dorsalis. OBLI. = Obliquus inferioris. OBLS. = Obliquus superioris. OBLV. = Obliquus ventralis. PHCE. = Pharyngoclavicularis externus. PHCI. = Pharyngoclavicularis internus. PRHY. = Protractor hyoidei. RAP. = Retractor arcus palatini. RCOMM. = Rectus communis. RECTV. = Rectus ventralis. SANT. = Supracarinales anterior. SO. = Sphincter oesophagi. STB. = Sternobranchialis. STH. = Sternohyoideus. TRD. = Transversus dorsalis.

traciins and balistids as both taxonomic and functional outgroups (Watrous and Wheeler, 1981).

Finally, Tyler (1980:Fig. 146) placed the Eocene fossil, *Proaracana*, as the sister group to the Aracaninae. In all but one of its discernable characters (mostly external features because of the excellent preservation of the carapace), it

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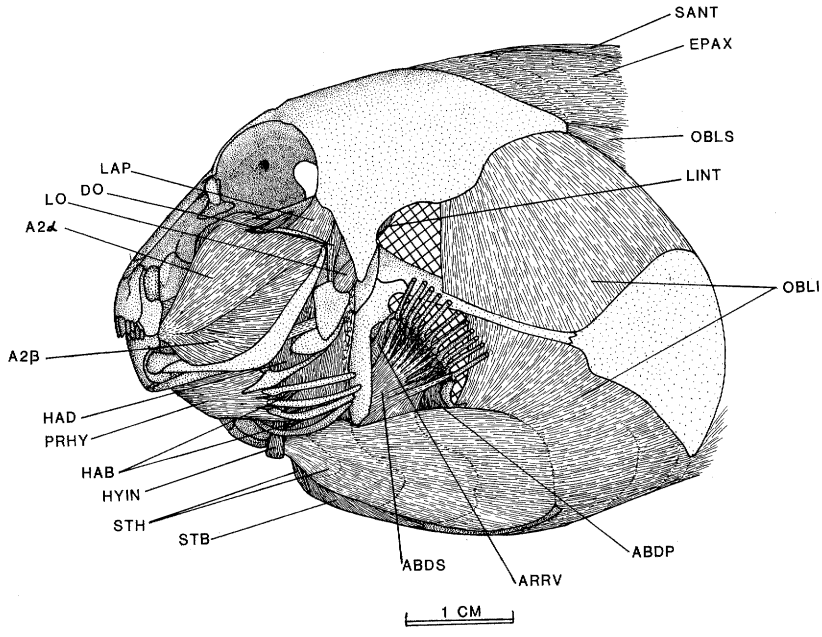


Fig. 8. *Aracana aurita*, left lateral view of superficial anterior musculature. See Fig. 7 for abbreviations.

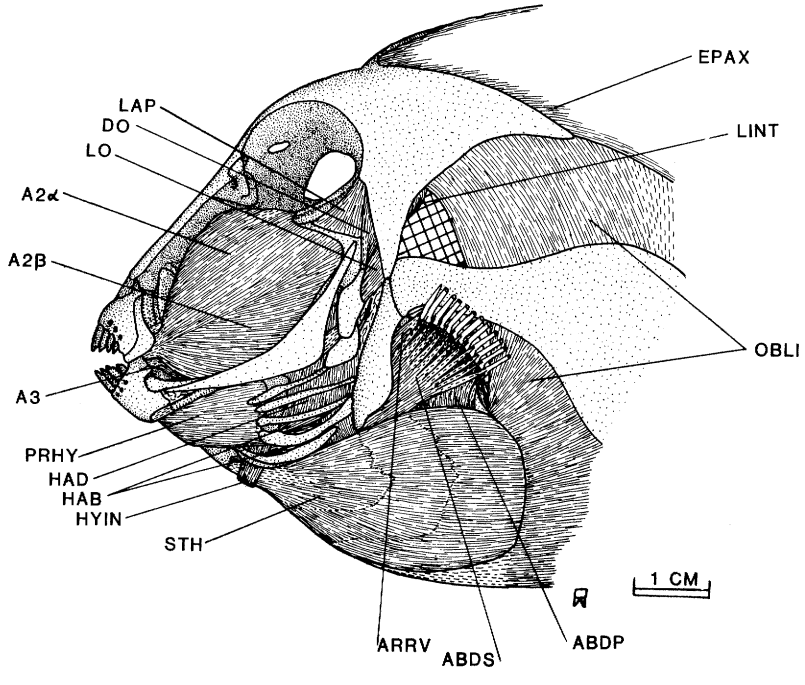


Fig. 9. *Strophiurichthys robustus*, left lateral view of superficial anterior musculature. See Fig. 7 for abbreviations.

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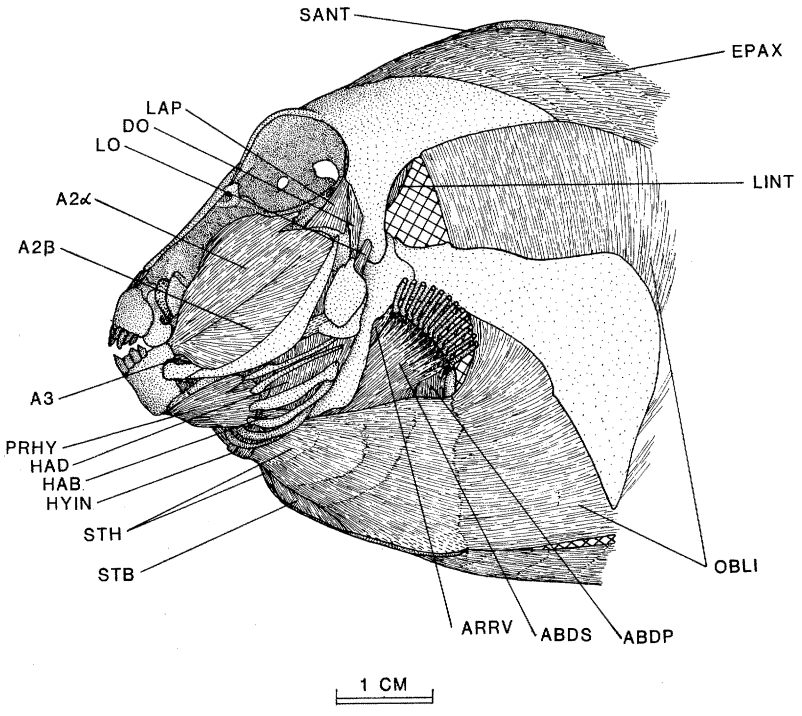


Fig. 10. *Anoplocapros amygdaloides,* left lateral view of superficial anterior musculature. See Fig. 7 for abbreviations.

agrees with the aracinins. However, all of these characters we consider to be primitive conditions (e.g., isolated scale plates on the caudal peduncle, deep body with central keels) that would be predicted to occur in the primitive sister group to the aracanins, ostraciins and ostraciids. Most aracinins have 11 caudal-fin rays (plesiomorphic) (The presence of either 10 or 11 caudal-fin rays in two species of *Kentrocapros* [Matsuura and Yamakawa, in press] may alter this argument, but the specimens have not been available to us.), while all ostraciins have 10 caudal-fin rays (apomorphic). The presence of 10 caudal-fin rays in *Proaracana* thus suggests that it belongs in the Ostraciinae and not the Aracaninae, and it is here placed in the former subfamily.

External Characters

Caudal peduncle scale plates.—It has generally been assumed (Fraser-Brunner, 1935; Fujii and Uyeno, 1979; Tyler, 1980) that the common ancestor of ostraciids had numerous, small, discrete scales on the caudal peduncle, and that this therefore represents the hypothesized plesiomorphous condition. This condition was thought to have been derived from a situation similar to that found in Recent balistins where the peduncle is covered with relatively large, overlapping scales (but not as in monacanthins, where most of the scales have small bases which overlap irregularly, if at all). The character states of the peduncular plates as found in aracanins are discussed below.

Polyplacapros has the peduncle covered with discrete, non-overlapping scales (Fig. 4), some of which may fuse into fewer, larger scale plates (Fujii and Uyeno, 1979: Figs. 7A, B). In the latter case, the squamation ends posteriorly with a pair of prominent saddles, one centered on the dorsal midline and one on the ventral midline, both of which taper as they near the lateral midline and almost meet one another. These saddles of composite plates thus appear triangular in lateral view. *Kentrocapros* has a variety of plate patterns on the peduncle. There may be a series of four small saddles dorsally and ventrally, with incomplete fusion between the individual scales making up each saddle, as in

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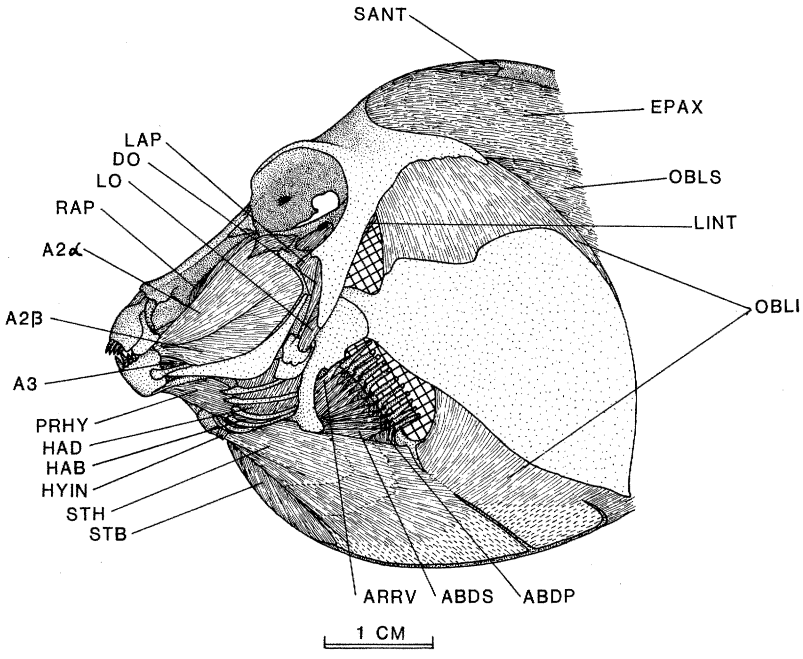


Fig. 11. *Caprichthys gymnura*, left lateral view of superficial anterior musculature. See Fig. 7 for abbreviations.

Kentrocapros aculeatus (Fig. 5). In other species, such as *K. rosapinto*, there is a mosaic of scales on the anterodorsal and anteroventral regions of the peduncle which taper into single scale plates posteriorly (K. Matsuura, pers. comm.). *Aracana* possesses a posterior pair of triangular saddles as described for some specimens of *Polyplacapros*, a similar but smaller pair of saddles anterior to this, and several isolated scattered scales or scale plates elsewhere on the peduncle. *Strophiurichthys* is similar to *Aracana*, but possesses only two pairs of triangular saddles. In *Anoplocapros*, the anterior pair of saddles is as described above, but posteriorly there is a complete bony ring around the caudal peduncle. This ring is also present in *Capropygia*, where the anterior pair of saddles apparently is represented by two smaller pairs of similar saddles. Finally, in *Caprichthys*, all that remains of peduncular scales or scale plates are two to three very small isolated scales ventrally and a small scale plate and an isolated scale dorsally. These lie in the same position as the anterior pair of saddles of other genera, and there is no trace of a bony ring or scales posterior to this (Tyler, 1980: Figs. 130–134 provides lateral views of the form of the carapace of the last five genera).

In ostraciins, no independent scale plates or scales are present on the caudal peduncle. Since independent scales cover the peduncle of at least the generalized members of the ostraciid sister group (the balistids) and since at least some aracanins possess peduncular scales, we conclude that reduction of scales and/or their consolidation into scale plates represent specializations. We have provided evidence to show that both the aracanins and ostraciins are monophyletic taxa, and therefore the scale-less peduncle of ostraciins and the almost scale-less peduncle of *Caprichthys* are not due to common ancestry. We are unable to interpret the nature of the scales and scale plates of *Polyplacapros* and *Kentrocapros* other than to suggest that, because scale coverage is more extensive and includes many individual scales, it represents the most primitive condition for aracanins. Reduction and consolidation of scale plates appear to follow a sequential transformation series as follows: *Aracana*, *Strophiurichthys*, *Anoplocapros*, *Capropygia* and *Caprichthys*. Loss of the independent scales present in *Aracana* leads to a condition like that found in *Strophiurichthys*; fusion of the posterior pair of saddles in the midlateral region results in the condition seen in *Anoplocapros*; break-up

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and reduction of the anterior pair of saddles results in the arrangement found in *Capropygia*; and loss of the posterior ring gives rise to the configuration present in *Caprichthys*. It is possible that *Caprichthys* may fit in elsewhere, but any other relationship would then involve parallel reductions in two lineages.

Carapace ridges.—Some aracanins and all ostraciins lack a mid-ventral ridge, a condition that therefore must be considered plesiomorphous within the ostraciids. The presence of a sharply defined ridge in *Anoplocapros, Aracana, Caprichthys, Capropygia* and *Strophiurichthys* (Tyler, 1980: Figs. 130–134) indicates that this group of five genera is monophyletic. *Kentrocapros* (Fig. 5) and *Polyplacapros* (Fig. 1) have a gently rounded ridge or none at all (Fujii and Uyeno, 1979: Fig. 3, Table 2). Further, most aracanins and ostraciins possess a mid-dorsal carapace ridge which begins above the eye and ends before the origin of the dorsal fin. This must, therefore, represent the plesiomorphous condition because the aracanins and ostraciins have been shown each to be monophyletic and the sister group of each other. The dorsum of *Aracana, Kentrocapros* (Fig. 5) and *Polyplacapros* (Fig. 1), in contrast, is flat or gently rounded. We identify this as a synapomorphy for these three taxa, and conclude that the morphologically similar condition in the ostraciin *Ostracion* is due to convergence, and is also an autapomorphy of *Ostracion*.

Rigidity of carapace.—In ostraciins and most aracanins, the carapace forms a rigid "box" except in the region of the gill opening, and in some cases on the cheeks. The carapace is markedly less rigid in the ventral region of the belly in *Kentrocapros* and *Polyplacapros* (especially in the latter), a condition we interpret to be specialized. We are aware that most arguments previously advanced would put exactly the opposite interpretation on this character distribution, since the primitive condition of the carapace in the ancestral ostraciid(s) must at one stage have been less consolidated than it is in Recent species. However, the fact that all ostraciins (the outgroup) and most aracanins have a rigid carapace in the mid-ventral region requires that this was the condition in the common aracanin-ostraciin ancestor, and that loss of rigidity is a specialization (this transformation thus involves a single change in character state). Conversely, if the immediate common ancestor of these two taxa possessed a more flexible mid-ventral region (retained, in this argument, by *Polyplacapros* and to a somewhat lesser extent by *Kentrocapros*) then the remaining aracanin genera as well as the ostraciins have independently acquired a more rigid mid-ventral region. Since the latter involves two changes in character state, we prefer the first hypothesis (especially since there are no ostraciins with a mid-ventrally flexible carapace).

Osteological Characters

Parasphenoid.—The ventral margin of the parasphenoid anterior to the orbit is not laterally expanded in balistids and in the aracanin genera *Anoplocapros, Kentrocapros, Polyplacapros* (Fig. 6a, b; the expanded parasphenoid cannot be seen in a dorsal view of the skull, Fig. 6c) and *Strophiurichthys*. In *Aracana*, the posterior region near the prootic shelves is laterally expanded to form a partial "hard palate" over the rear of the oral cavity. In *Caprichthys* and *Capropygia* this portion of the parasphenoid is unexpanded, but the anterior portion behind the vomer is laterally expanded to form a "hard palate" over the front of the oral cavity (for the latter two conditions see Tyler, 1980: Fig. 144). In ostraciins, the entire ventral edge of the parasphenoid is laterally expanded to form a "hard palate" over the entire oral cavity (Tyler, 1980: Figs. 150, 151). The following discussion is based on the assumption that both the aracanins and ostraciins are monophyletic, and are sister groups (as documented earlier in this paper).

The problem of the interpretation of these various character states revolves around the plesiomorphous condition of the parasphenoid in the ancestral ostraciid. If we assume that it was unexpanded (as in balistids and some aracanins) then: a) *Aracana* is autapomorphic; b) *Caprichthys* and *Capropygia* are sister groups; and c) the ostraciins are monophyletic. Thus, the conclusions in "a" or "b" would not represent an early stage in a transformation series leading to the ostraciin condition. If, however, the common ancestor of the ostraciids possessed a full "hard palate," as in ostraciins, secondary loss has occurred in at least two separate ways: a) *Aracana* lost the expansion of the parasphenoid anteriorly; b) and the common ancestor of *Caprichthys* and *Capropygia* lost the expansion of the parasphenoid posteriorly. Given these two alternatives, we prefer the first as being the most probable in terms of outgroup comparisons, and

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conclude that *Aracana* is autapomorphic and that *Caprichthys* and *Capropygia* are sister groups.

Hemal canal.—The hemal canal of most aracanins and ostraciins enters the fused centrum-epural-hypural-parhypural plate and exits bilaterally from the plate through a foramen. This foramen presumably represents the separation between the parhypural and lower hypural as it occurs in teleosts with a generalized caudal fin skeleton. In three aracanin genera, (*Anoplocapros, Capropygia* and *Caprichthys*) no foramen is detectable, and this condition is considered to represent a synapomorphy linking these three genera. Parenthetically, we note that in at least some ostraciins (e.g., *Ostracion*), the hemal canal does not even enter the caudal plate. We suggest that it is unparsimonious to regard this character as linking some ostraciins with some aracanins, in the light of our analysis in the preceding section, and we conclude with confidence that this resemblance between some aracanins and some ostraciins is convergent.

Supraneural (=*basal pterygiophores of spiny dorsal fin*).—Aracanins possess a long supraneural extending far anteriorly from its posterior contact with the first basal pterygiophore of the dorsal fin, but not reaching the posterior margin of the skull (Fig. 2). Ostraciins, on the other hand, have a short, little developed supraneural attached to the first basal pterygiophore of the dorsal fin (Tyler, 1980: Fig. 169). If our identification of the supraneural as an element representing one or more basal pterygiophores of a now-absent spiny dorsal fin is correct, then the condition in the second outgroup, the balistids, seems autapomorphic, for in these fishes the pterygiophores are attached anteriorly to the skull. Further searching for information pertinent to a resolution of character state polarity depends on what is assumed to be the sister group of the balistoids (=balistids + ostraciids). As stated in the Introduction, Winterbottom's (1974a) cladistic analysis placed the tetraodontoids in this position, while Tyler's (1980:4) gradistic study cast the triacanthoids in this role. Given that the Eocene *Eoplectus*, with its beak-like jaws and consolidated teeth, is geneologically a tetraodontoid, a comparison can be made between the balistoid basal pterygiophores/supraneural and those of the spiny dorsal fin of *Eoplectus* and triacanthids. In *Eoplectus*, the first two pterygiophores bear three spines, and the third pterygiophore has an anterodorsal to posteroventral inclination. Loss of the three posterior spines and narrowing of the third pterygiophore would produce a condition much like a balistid first dorsal fin (Tyler, 1980: Figs. 17, 51). On the other hand, posterior migration of the pterygiophores and reduction in size, number and spine development would produce a situation similar to that found in the primitive tetraodontoid *Triodon* (triodontids), with further reduction resulting in a form of supraneural like that found in tetraodontids, diodontids and molids. The supraneural of ostraciins is anatomically identical to that of the latter three families. However, ostraciins do not have the tetraodontoid condition of beak-like jaws and they do possess the autapomorphies of balistoids listed in the preceding section. We have considered the dorsal-fin spines of *Eoplectus* to be a primitive condition for tetraodontoids, and we conclude that the look-alike supraneural of ostraciids and the tetraodontids, diodontids and molids can only be due to convergence.

Considering the possibility that the triacanthids are the sister group of the balistoids, the balistid-type spiny dorsal fin could be derived from a triacanthoid-like condition just as easily as from an *Eoplectus*-like structure (particularly if Rath's, 1859, description and figure of the Oligocene *Cryptobalistes* are accurate—although we suspect they are not). We conclude from the above that the short supraneural is an autapomorphy for ostraciins. Within aracanins, the supraneural either has a deep, ventral flange (depth 3–6 times in length in *Anoplocapros, Capropygia, Caprichthys, Strophiurichthys*), or is thin and narrow (depth 8.5–11 times in length in *Aracana, Kentrocapros, Polyplacapros*). In generalized balistids, the pterygiophores of the spiny dorsal fin have a deep ventral flange which we therefore interpret as the primitive condition for ostraciids. The narrow flange thus is evidence of reduction, indicating that *Aracana, Kentrocapros* and *Polyplacapros* form a monophyletic assemblage.

Dorsal and anal fin rays and pterygiophores.—All ostraciins and most aracanins possess 9–11 dorsal and anal-fin rays, and this would appear to be the primitive condition for these fishes (Fig. 2). *Capropygia* and *Caprichthys* usually have 13 dorsal-fin rays and 12 anal-fin rays, indicating that these two genera are sister groups. A similar supernumerary condition exists for the dorsal and anal pterygiophores, but we consider

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this part of the same character complex and do not use it separately.

Myological Characters

Adductor mandibulae.—Balistins, ostraciins, *Kentrocapros* (Winterbottom, 1974a: Fig. 115) and *Polyplacapros* (Fig. 7a) have the A2 section of this muscle completely subdivided into two sections (α and B). In the remaining five aracanin genera, these two sections are fused posteriorly, indicating that these five genera form a monophyletic group (Figs. 8–11; Winterbottom, 1974a: Fig. 125 of *Capropygia*).

Sternobranchialis.—In *Aracana*, *Kentrocapros*, *Polyplacapros* (Fig. 7b) and *Strophiurichthys* the muscle fibers of the sternobranchialis form a single bundle which grades into an aponeurosis. An anterior and posterior tendon arise from the aponeurosis to insert on the hypo- and ceratobranchial elements of the gill arches. In *Anoplocapros*, the dorsal half of the muscle is separated into two sections, with the anterior half serving the anterior tendon and the posterior half giving rise to the posterior tendon. The fibers making up the ventral half of the muscle are not separated. *Caprichthys* and *Capropygia* have two separate muscle bundles (i.e., the separation is complete). Ostraciins also have two separate sections, of which the anterior is greatly enlarged, but balistids possess a single, undivided muscle serving two tendons as described for the four aracanin genera above. We conclude that the primitive ostraciid condition is for a single undivided muscle, and that *Anoplocapros* is the sister group of the group composed of the *Caprichthys–Capropygia* lineage. This implies that the condition in ostraciins is the result of convergence. We accept that we can be accused of vacillation here, for we rejected an argument similar to the above in the case of the carapace rigidity (q.v.). However, one can reverse the above argument and state that the primitive condition of the sternobranchialis in ostraciids consisted of two separated muscles (as in some aracanins and all ostraciins). This would mean that these two muscles must have reversed evolutionary direction: i.e., that fusion has occurred in those aracanins with a single bundle of muscle fibers. In all cases known to us where fusion of separate muscles into a single entity has occurred in teleosts, the site of fusion is demarcated by a myocomma, or by complex pinnation of the fibers (e.g., protractor hyoidei of teleosts, and the fusion of A1 to A2 in most congrogadids, respectively). No such demarcation is evident in the aracanins examined, in spite of the fact that one of the genera (*Anoplocapros*) exhibits this "fusion" in an intermediate or incomplete stage and could thus be expected to show some sign of it.

Transversus ventralis IV.—The muscle is relatively well developed (albeit smaller than the transversus ventralis V) in *Kentrocapros*, *Polyplacapros* and the ostraciins, but weak and poorly developed in *Anoplocapros*, *Aracana*, *Caprichthys*, *Capropygia* and *Strophiurichthys*, indicating that the group formed by the latter five genera is monophyletic.

Levator externus III.—In ostraciins and most aracanins, the third external levator is partly fused to the fourth levator dorsally, but remains separate ventrally. In *Kentrocapros* and *Polyplacapros* (Fig. 7b) these fibers form a single muscle bundle, with the tendon to epibranchial 3 arising from the anteroventral surface of the combined third and fourth external levators. We conclude that this latter condition represents a synapomorphy linking these two taxa.

The remaining musculature provided no phylogenetic information that we were able to analyse.

Discussion

Analysis of the character state trees for aracanins produces a phylogeny of the genera as shown in Fig. 12. Two of the characters analysed contradict part of this hypothesis and, perhaps significantly, both indicate the same alternative relationships. These characters are the narrow supraneural and the absence of a dorsal carapace ridge, both of which indicate that *Aracana*, *Kentrocapros* and *Polyplacapros* share a common ancestor not shared by other aracanin genera. Reversal of the assumptions about these character states would, of course, corroborate the monophyly of the group made up of *Anoplocapros*, *Caprichthys*, *Capropygia* and *Strophiurichthys*. With regard to the absence of a dorsal carapace ridge, we note that both the Ostraciinae and Aracaninae include members with and without such a ridge. If, in fact, a flat dorsum is primitive for ostraciids, then in addition to corroboration of our preferred hypothesis, it implies that *Ostracion* is the primitive sister group to all other ostraciins (which are thus mono-

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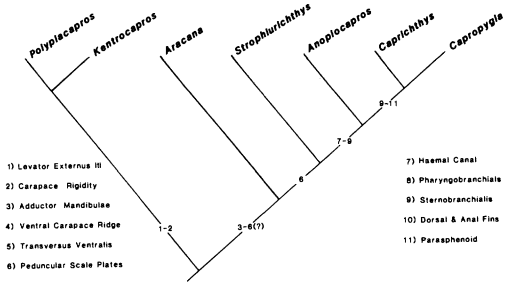


Fig. 12. Phylogeny of the genera of the Aracaninae. Numbers refer to characters as discussed in the Analysis section of this paper.

phyletic). We note that, in contrast, presence or absence of a ventral carapace ridge is not subject to such a reinterpretation, since it is absent in all ostraciins and some aracanins. It is this character (present = apomorphic) along with two and possibly three others (Fig. 12) which suggests that *Anoplocapros, Aracana, Caprichthys, Capropygia* and *Strophiurichthys* form a monophyletic assemblage. The character states for the supraneural bone were extensively discussed above, and although such meager evidence as we were able to glean from outgroups indicates that a deep supraneural is primitive, we are quite prepared to learn that our assignment of polarity is incorrect.

We therefore conclude that the two characters that contradict our hypothesis of relationships (Fig. 12) represent either incorrect assignment of character polarity or homoplasy. We are unable to decide between these two interpretations.

The most parsimonious hypothesis thus corroborates one of the two contradictory hypotheses proposed by Fujii and Uyeno (1979): that *Polyplacapros* is more closely related to *Kentrocapros* than to any other aracanin. Our analysis does not support their contention that most of the characters by which *Polyplacapros* differs from other aracanins or shares with *Kentrocapros* are primitive for aracanins.

Tyler's (1980: Fig. 146) phylogeny of aracanin genera (exclusive of *Polyplacapros*) presents a very different picture from that developed here. In his analysis, *Strophiurichthys* is the sister group to all other genera, with *Kentrocapros* and *Aracana* forming one lineage, and *Anoplocapros, Caprichthys* and *Capropygia* being sequentially related to each other in the second lineage. Tyler's (1980) evidence is based mainly on plesiomorphous character states and therefore is inadmissible in a cladistic interpretation.

Three of the seven aracanin genera appear to be monotypic (*Caprichthys gymnura, Capropygia unistriata* and *Polyplacapros tyleri*), two genera apparently contain two species each (*Aracana aurita, A. ornata*; *Strophiurichthys inermis, S. robustus*) and two genera, *Anoplocapros* and *Kentrocapros* have three species (*A. amygdaloides, A. grayi* and *A. lenticularis*; *K. aculeatus, K. flavofasciatus* and *K. rosapinto*). The scarcity of aracanin specimens in museum collections, coupled with allometric growth and sexual dimorphism (Fraser-Brunner, 1940) makes even alpha-level systematics of the group difficult. Furthermore, we are aware of a few other new species of aracanins awaiting description (T. Uyeno, pers. comm.). We are thus reluctant to tamper with the present finely split generic allocations in spite of our feeling that the Aracaninae is overstocked with genera.

In order to reflect our phylogeny in a classification we have adopted three conventions. The first is to split the Aracaninae into two tribes, the Aracanini (*Aracana, Strophiurichthys, Anoplocapros, Caprichthys* and *Capropygia*) and the Kentrocaprini (*Kentrocapros* and *Polyplacapros*). The type-species of the tribes are *Aracana aurita* (Shaw) and *Kentrocapros aculeatus* (Houttuyn), respectively. Since only two genera are included in the Kentrocaprini, their sister group status is obvious. Within the Aracanini, two methods of presenting the phylogenetic information in the classification can be used. Sequencing (e.g., Nelson, 1974; Patterson and Rosen, 1977) will reveal the proposed phylogeny from the classification until such time as one or more of the three "basal" genera in question (*Aracana, Strophiurichthys* or *Anoplocapros*) is demonstrated to share a sister group status with a hypothetical undescribed genus to the exclusion of the other genera of the Aracanini. Alternatively, additional information could lead to a refutation of the hypotheses erected here, and lead to classificatory instability. The numerical suffix system described and used by Winterbottom (1974a:98) requires a little more background, but is easily converted to accommodate subsequent discoveries and changes. We use both systems (since they are, in this case, compatible) in the classification of the Aracanini below:

Subfamily: Aracaninae
Tribe: Kentrocaprini
Genus:*Kentrocapros*
Polyplacapros

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Tribe: Aracanini
Genus:*Aracana* (1)
Strophiurichthys (2)
Anoplocapros (3)
Caprichthys (4)
Capropygia (4)

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Literature Cited

Fraser-Brunner, A. 1935. Notes on the plectognath fishes.-II. A synopsis of the genera of the family Ostraciontidae. Ann. Mag. Nat. Hist., ser. 10 16: 313–320.

———. 1940. Notes on the plectognath fishes.-IV. Sexual dimorphism in the family Ostraciontidae. *Ibid.*, ser. 11 6:390–392.

———. 1941. Notes on the plectognath fishes.-VII. The Aracanidae, a distinct family of ostraciontoid fishes, with descriptions of two new species. *Ibid.*, ser. 11 8:306–313.

Fujii, E., and T. Uyeno. 1979. *Polyplacapros tyleri*, a new genus and species of ostraciid trunkfish from off eastern Australia and Norfolk Ridge. Jap. J. Ichthyol. 26:1–10.

Hennig, W. 1966. Phylogenetic systematics. Urbana, Univ. Illinois Press.

Matsuura, K. 1981. *Xenobalistes tumidipectoris*, a new genus and species of triggerfish (Tetraodontiformes, Balistidae) from the Marianas Islands. Bull. Natl. Sci. Mus., ser. A 7:191–200.

Nelson, G. J. 1974. Classification as an expression of phylogenetic relationships. Syst. Zool. 22:344–359.

Patterson, C., and D. E. Rosen. 1977. Review of ichthyodectiform and other Mesozoic teleost fishes and the theory and practice of classifying fossils. Bull. Amer. Mus. Nat. Hist. 158:81–172.

Smith, J. L. B. 1949. A new aracanid fish from South Africa. Ann. Mag. Nat. Hist., ser. 12 2:354–359.

Tyler, J. C. 1962. The pelvis and pelvic fin of plectognath fishes; a study in reduction. Proc. Acad. Nat. Sci. Phila. 114:207–250.

———. 1968. A monograph on plectognath fishes of the Superfamily Triacanthoidea. Monogr. Acad. Nat. Sci. Phila. 16:1–364.

———. 1980. Osteology, phylogeny and higher classification of the fishes of the order Plectognathi (Tetraodontiformes). NOAA Techn. Rept. NMFS Circ. 434.

Watrous, L. E., and Q. D. Wheeler. 1981. The out-group comparison method of character analysis. Syst. Zool. 30:1–11.

Winterbottom, R. 1974a. The familial phylogeny of the Tetraodontiformes (Acanthopterygii, Pisces) as evidenced by their comparative myology. Smith. Contrib. Zool. 155:1–201.

———. 1974b. A descriptive synonymy of the striated muscles of the Teleostei. Proc. Acad. Nat. Sci. Phila. 125:225–317.

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